

Mobile Application Development

Lecture 05

Today's Agenda

- Unit of Measurement
- Types of Layouts
 - View & View Groups
 - Linear Layout
 - Relative Layout
 - Constraints Layout

Units of Measurement

Sr.No	Unit & description
1	dp Density-independent pixel. 1 dp is equivalent to one pixel on a 160 dpi (dot per inch) screen.
2	sp Scale-independent pixel. This is similar to dp and is recommended for specifying font sizes
3	pt Point. A point is defined to be 1/72 of an inch, based on the physical screen size.
4	px Pixel. Corresponds to actual pixels on the screen

View

- This class represents the basic building block for user interface components.
- A View occupies a rectangular area on the screen and is responsible for drawing and event handling.
- View is the base class for *widgets*, which are used to create interactive UI components (buttons, text fields, etc.).

Using View

- All of the views in a window are arranged in a single tree.
- You can add views either from code or by specifying a tree of views in one or more XML layout files.
- There are many specialized subclasses of views that act as controls or are capable of displaying text, images, or other content.

ViewGroup

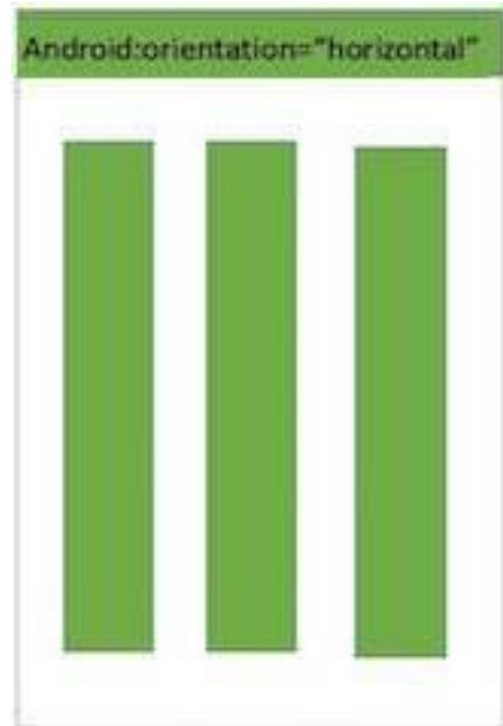
- A ViewGroup is a special view that can contain other views (called children)
- The view group is the base class for layouts and views containers
- This class also defines the ViewGroup.LayoutParams class which serves as the base class for layouts parameters
- ViewGroup.LayoutParams class which serves as the base class for layouts parameters.
- The base LayoutParams class just describes how big the view wants to be for both width and height. For each dimension, it can specify one of:
 - FILL_PARENT (renamed MATCH_PARENT in API Level 8 and higher), which means that the view wants to be as big as its parent (minus padding)
 - WRAP_CONTENT, which means that the view wants to be just big enough to enclose its content (plus padding)

Common Attributes of View & View Groups

Sr.No	View & description
1	layout_width Specifies the width of the View or ViewGroup
2	layout_height Specifies the height of the View or ViewGroup
3	layout_marginTop Specifies extra space on the top side of the View or ViewGroup
4	layout_marginBottom Specifies extra space on the bottom side of the View or ViewGroup
5	layout_marginLeft Specifies extra space on the left side of the View or ViewGroup
6	layout_marginRight Specifies extra space on the right side of the View or ViewGroup
7	layout_gravity Specifies how child Views are positioned
8	layout_weight Specifies how much of the extra space in the layout should be allocated to the View

Linear Layout

- *Android LinearLayout is a view group that aligns all children in either vertically or horizontally.*



Linear Layout Attributes

- Following are the important attributes specific to LinearLayout

Sr.No	Attribute & Description
1	android:gravity This specifies how an object should position its content, on both the X and Y axes. Possible values are top, bottom, left, right, center, center_vertical, center_horizontal etc.
2	android:orientation This specifies the direction of arrangement and you will use "horizontal" for a row, "vertical" for a column. The default is horizontal.
3	android:weightSum Sum up of child weight

Linear Layout Example

Username

Email

Password

☒ Male ☐ Female

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